

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2455427.94306 +/- 0.00074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.07 +/- 0.052
Transit depth (Rp/Rs)^2	0.48 +/- 0.72 [%]
Orbital Inclination (inc)	87.1 +/- 1.94
Airmass coefficient 1 (a1)	1.073 +/- 0.019
Airmass coefficient 2 (a2)	-0.071 +/- 0.095
Scatter in the residuals of the lightcurve fit is	1.77 %
Best Comparison Star	#2 - [111, 423]
Optimal Aperture	3.1
Optimal Annulus	9.65
Transit Duration (day)	0.05 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.07 ± 0.052 vs 0.1592 (deviation 1.72σ)
Duration fit vs published	0.05 d vs 0.0946 d (deviation 2.35σ)
Mid-time O–C	1.3 min
Depth SNR	1.3
Residual scatter	1.77 %

Light curve

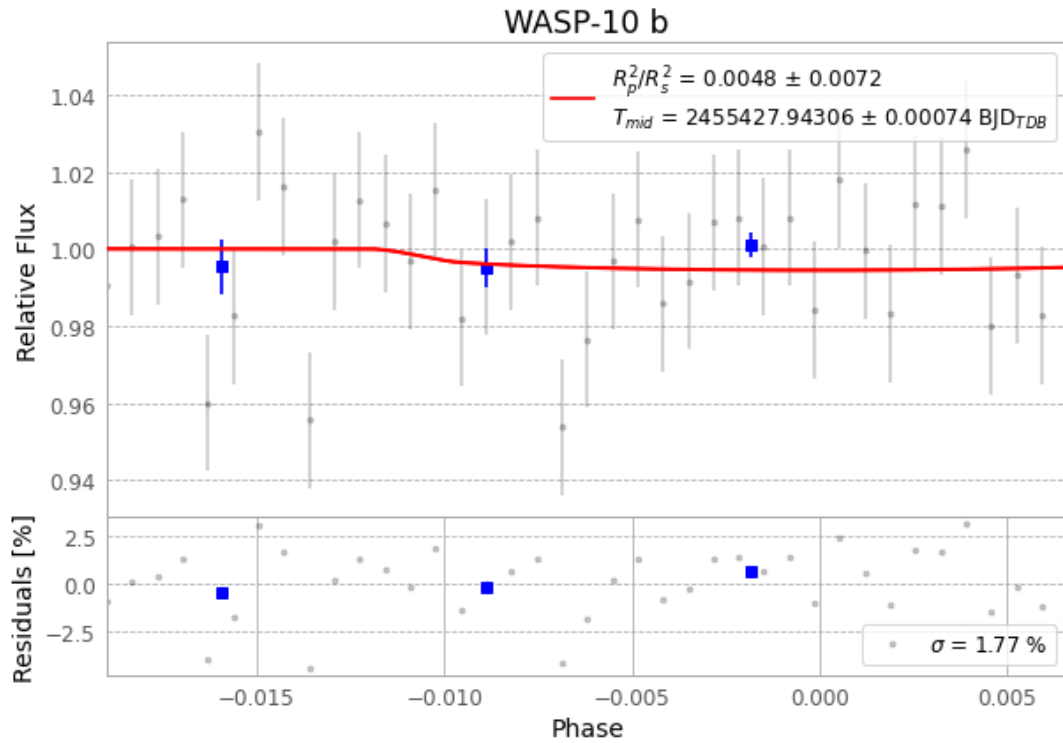


Figure 1 — FinalLightCurve_WASP-10 b_2010-08-19.png (SHA-256 aea8f4f1fa007565...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [232, 203], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5de167ac176e889d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

39 FITS frames (26 MB), WASP-10100819090614.FITS → WASP-10100819110014.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-100819.log (3c48fb6cfae8...), FinalLightCurve_WASP-10 b_2010-08-19.png (aea8f4f1fa00...), FinalLightCurve_WASP-10 b_2010-08-19.csv (5e45eb83bf39...)

Compute resources

Wall time 92.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-21 23:19Z.